

Park 2016 Table-4 OR vs mean |SHAP| rank (migraine/park)

What this compares. Park et al. 2016 report odds ratios for self-reported migraine triggers in the SHD cohort - a population-level, same-day association between a trigger and a migraine day. The model's mean |SHAP| rank is a data-driven importance for *next-day* individual prediction from the park trigger features. The two answer different questions, so this is a convergence check, not a validation: a positive Spearman correlation means the model recovers Park's trigger ordering; a value near zero or negative means it weights the triggers differently, which is expected when same-day cross-sectional odds ratios are asked to drive a next-day forecast on a small feature set. Each panel shows the leaf's full provenance (architecture, hyperparameter-tuning, data-split ratio and split type) and the odds-ratio-rank vs SHAP-rank scatter against the agreement diagonal.

runner_up: stacked_2xgb_meta_lr (xgboost, mean |SHAP|)

stacked_2xgb_meta_lr [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.625, AUPRC 0.115, calib +0.82

Spearman correlation (SHAP rank vs Park OR rank): **-0.429** over 6 triggers.

trigger	Park OR	OR rank	SHAP rank	mean SHAP
travel_today	6.4	1	5	0.0088
hormonal_changes_today	3.5	2	2	0.0279
noise_today	2.8	3	6	0.0054
alcohol_today	2.5	4	4	0.0157
overeating_today	2.4	5	1	0.0292
stress_today	1.8	6	3	0.0205

